

Garry Nelson

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Electrical Engineering graduate, GPA 3.98/4.00 Summa Cum Laude, specializing in machine learning, deep learning, and real-time AI systems. Published IEEE Access (Q1) author with experience developing embedded AI systems, cloud-deployed inference APIs, and real-time sensing pipelines.

Education

Telkom University

Bandung, Indonesia

Bachelor of Electrical Engineering. **GPA: 3.98 / 4.00 – Summa Cum Laude**

September 2022 – January 2026

Relevant coursework: Machine Learning, Artificial Intelligence, Probability & Statistics, IoT Systems, Linear Algebra

Experience

PT. Labtech Penta International Batam

Batam, Indonesia

IoT & Digital Twin Software Intern

June 2025 – August 2025

- Contributed in developing a real-time Digital Twin system for an industrial HVAC training unit by building a Unity-based interface integrated with an MQTT broker on Google Cloud Platform, enabling live bidirectional synchronization between physical equipment and its virtual counterpart
- Reduced manual monitoring overhead by implemented real-time synchronization logic between physical systems and virtual interfaces for monitoring and control workflows

Sensing, Signal Processing and Learning for Spatial Intelligence Laboratory

Bandung, West Java, Indonesia

Research Assistant – Radar & Deep Learning

September 2025 – January 2026

- Built end-to-end deep learning pipelines for FMCW radar-based Human Activity Recognition by designing spatiotemporal feature extraction architectures, resulting in two peer-reviewed publications including an IEEE Access Q1 paper
- Accelerated real-time activity classification research by developing and benchmarking multiple temporal deep learning models on radar point cloud sequences

Skills & Interest

Programming & Frameworks: Python, C, SQL, JavaScript, Dart, Git, Flutter, React

Machine Learning & AI: TensorFlow, Keras, PyTorch, scikit-learn, Hugging Face Transformers, Time-Series Modeling, Computer Vision

MLOps & Backend: FastAPI, Docker, MLflow, GitHub Actions, CI/CD, Google Cloud Platform

IoT & Embedded Systems: ESP32, Raspberry Pi, Arduino, PCB Design, MQTT, BLE, PLC

Language: Indonesian (Native), English (Professional Working Proficiency), Chinese Mandarin (Conversational)

Projects

Real-Time Human Fall Detection System using Radar and Deep Learning

(github.com/notgarryy/real-time-fall-detection-system-with-TTSNet)

- Designed and trained a TTSNet-based deep learning architecture for spatiotemporal feature extraction on FMCW radar point cloud sequences, achieving 98% classification accuracy across multiple activity classes
- Built a real-time inference pipeline for processing live FMCW radar data with low-latency fall detection to ensure safety of user
- Extended the research into a Temporal Convolutional Network-based architecture variant, resulting in two peer-reviewed publications

Fall Detection Inference API

(github.com/notgarryy/fall-detection-api)

- Built a production-ready REST API using FastAPI for real-time FMCW radar fall detection inference with confidence scoring
- Implemented preprocessing pipelines including DBSCAN clustering, sliding window inference, and TTSNet model serving
- Containerized the application using Docker and deployed to a cloud environment, demonstrating end-to-end MLOps capability

Face Recognition-Based Smart Door Lock System

(github.com/notgarryy/facenet-face-detection)

- Implemented a face authentication system using FaceNet for feature extraction and identity matching, achieving 88.9% recognition accuracy
- Deployed on Raspberry Pi 4 for real-time processing and physical door lock actuation
- Addressed lighting inconsistency through targeted data augmentation, improving model robustness across varying environmental conditions

Portable Air Particulate Monitoring System

(github.com/notgarryy/dustify-app)

- Developed an IoT-based PM2.5/PM10 air quality monitoring device using ESP32 and Sensirion SPS30 particulate sensors for real-time environmental sensing
- Built a cross-platform mobile application using Flutter and integrated it with the embedded monitoring device for real-time data visualization and Air Quality Index (AQI) monitoring
- Implemented wireless communication between embedded hardware and the mobile application to support live sensor data transmission and monitoring functionality

Leadership & Activities

Basic Computing Laboratory

Bandung, West Java, Indonesia

Vice Assistant Coordinator & Teaching Assistant

Jul 2023 – Jun 2025

- Raised practical course outcomes above department average by mentoring undergraduate students in C programming, debugging, and algorithmic problem solving across multiple cohorts
- Improved lab operational consistency by coordinating assistant scheduling and activities, and contributed to restructuring teaching materials and workflows across multiple semesters

Electronics and Industrial Automation Laboratory

Bandung, West Java, Indonesia

Teaching Assistant

Jan 2024 – Dec 2025

- Improved student proficiency in embedded IoT systems by teaching wireless communication protocols — Bluetooth, BLE, Wi-Fi, MQTT, and Firebase RTDB — across multiple practical cohorts
- Accelerated student onboarding to industrial automation by instructing PLC fundamentals, sensor and actuator wiring, Ladder Logic programming, and Human Machine Interface (HMI) operation using OMRON hardware

Publications

1. **Spatiotemporal Feature Learning for Real-Time Human Fall Detection System Using TCNs and FMCW Radar Point Clouds** – IEEE Access (Q1), 2026
2. **Real-Time Fall Detection Using FMCW Radar Point Clouds with TTSNet Spatiotemporal Feature Learning** – ASSET (Q3, SINTA 1), 2026